



ETHICAL HACKING ASSIGNMENT REPORT

STUDENT INFORMATION

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ADMISSION NO: BSCCS/2024/43869

COURSE: ETHICAL HACKING / CYBERSECURITY

TOOL USED: NMAP

PLATFORM: PARROT OS

DATE: 3RD JUNE 2026

LECTURER: MICHAEL NYORO

YEAR: 4 SEM: 2

PROJECT DETAILS

Project Title: Web Vulnerability Scanning and Foot printing Analysis

Selected Technologies: Kali Linux, Nmap

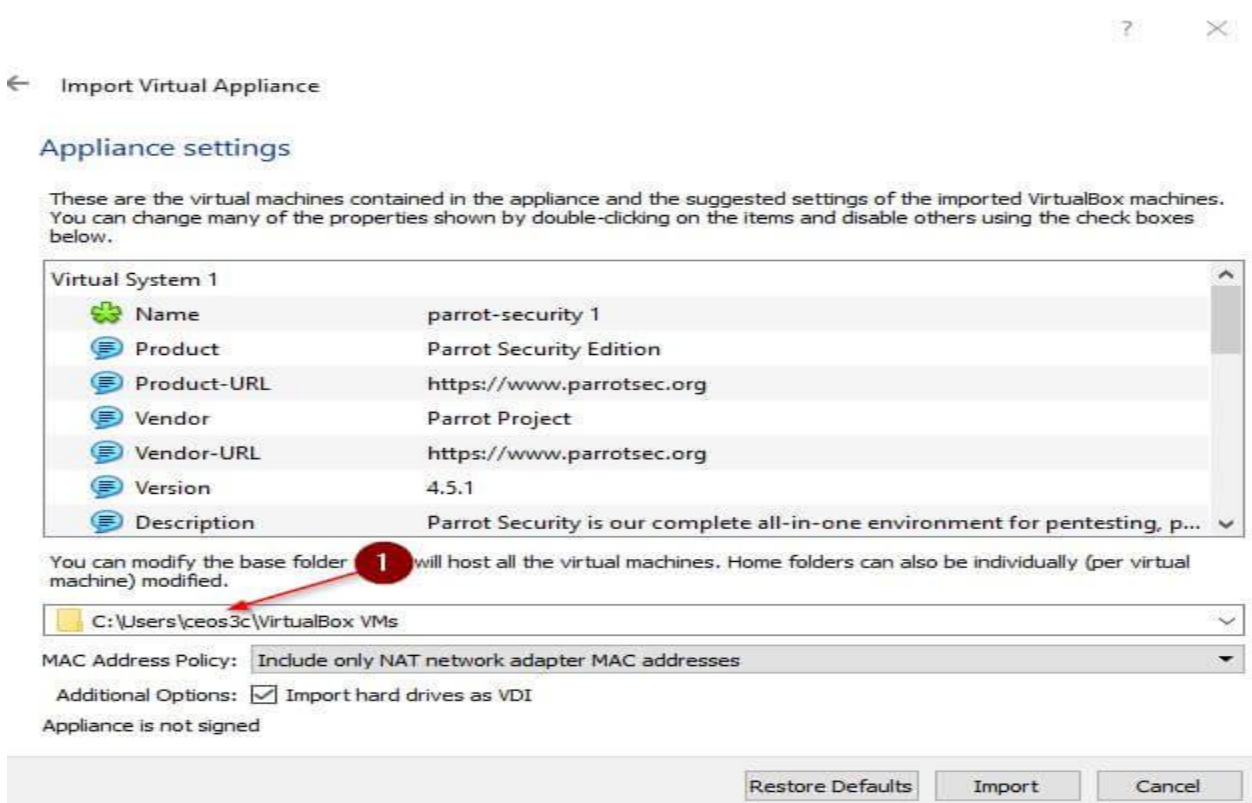
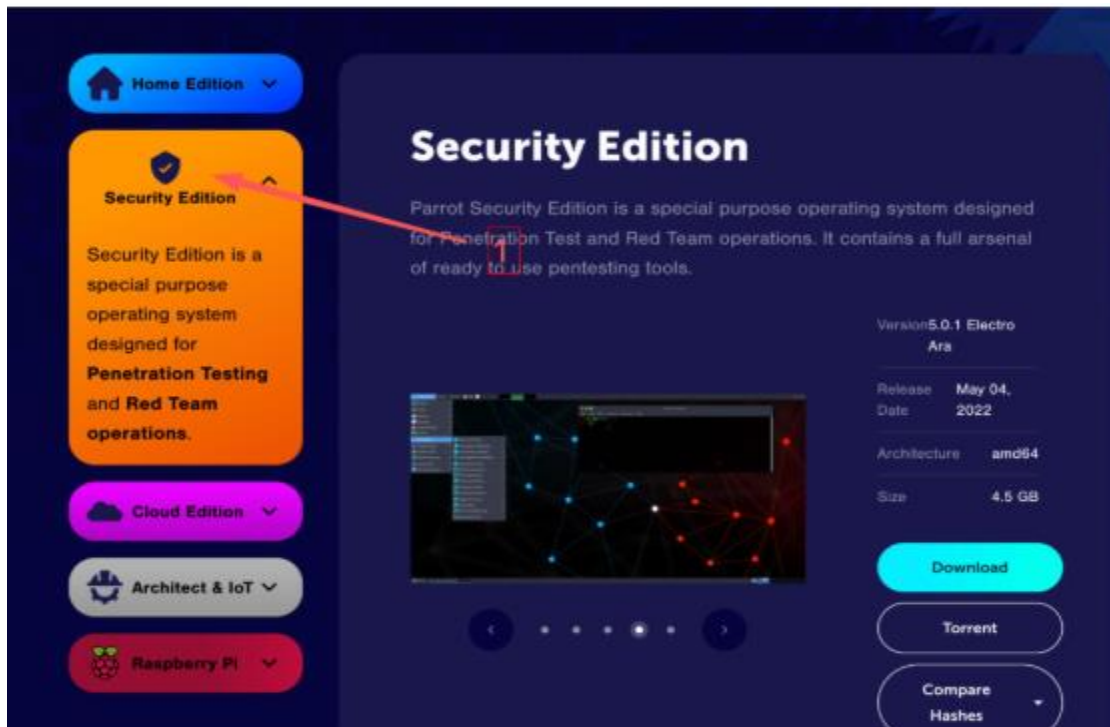
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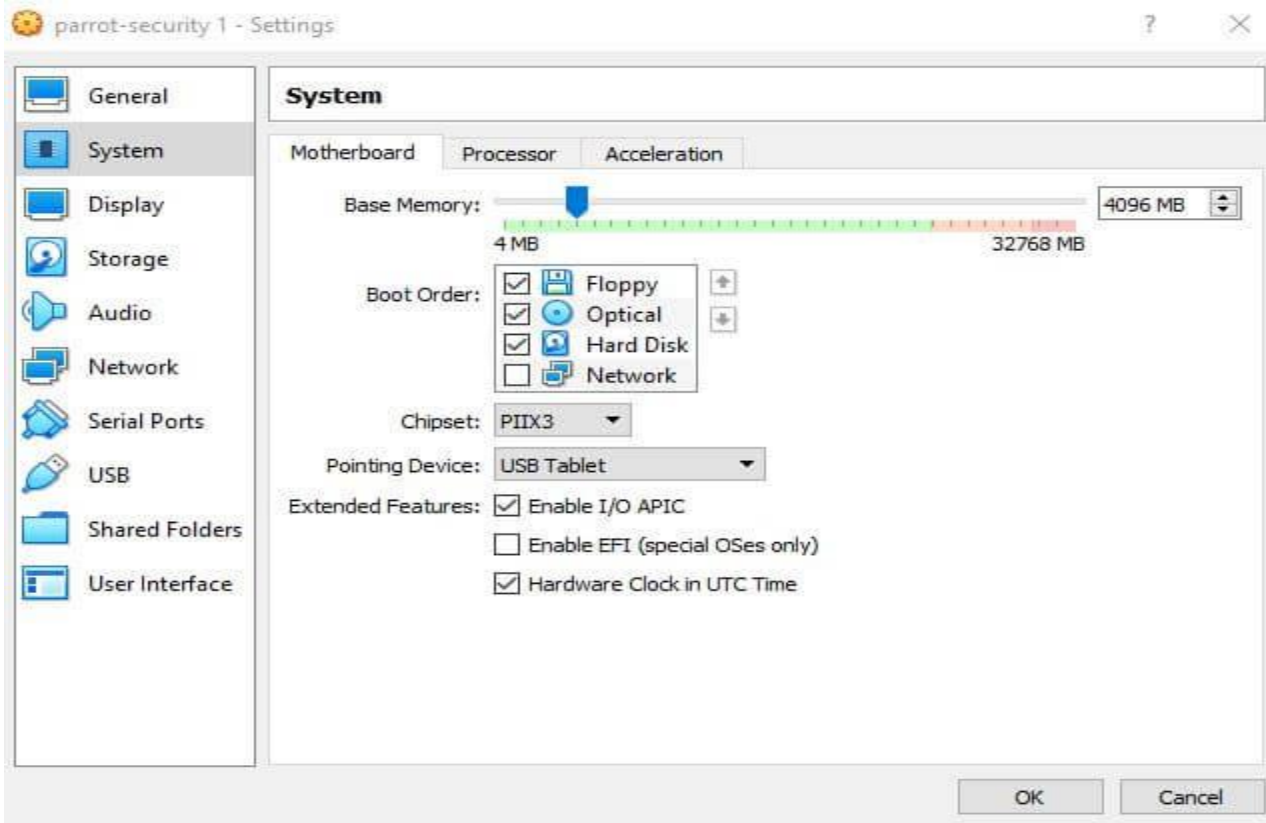
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ETHICAL HACKING AND COUNTER MEASURES

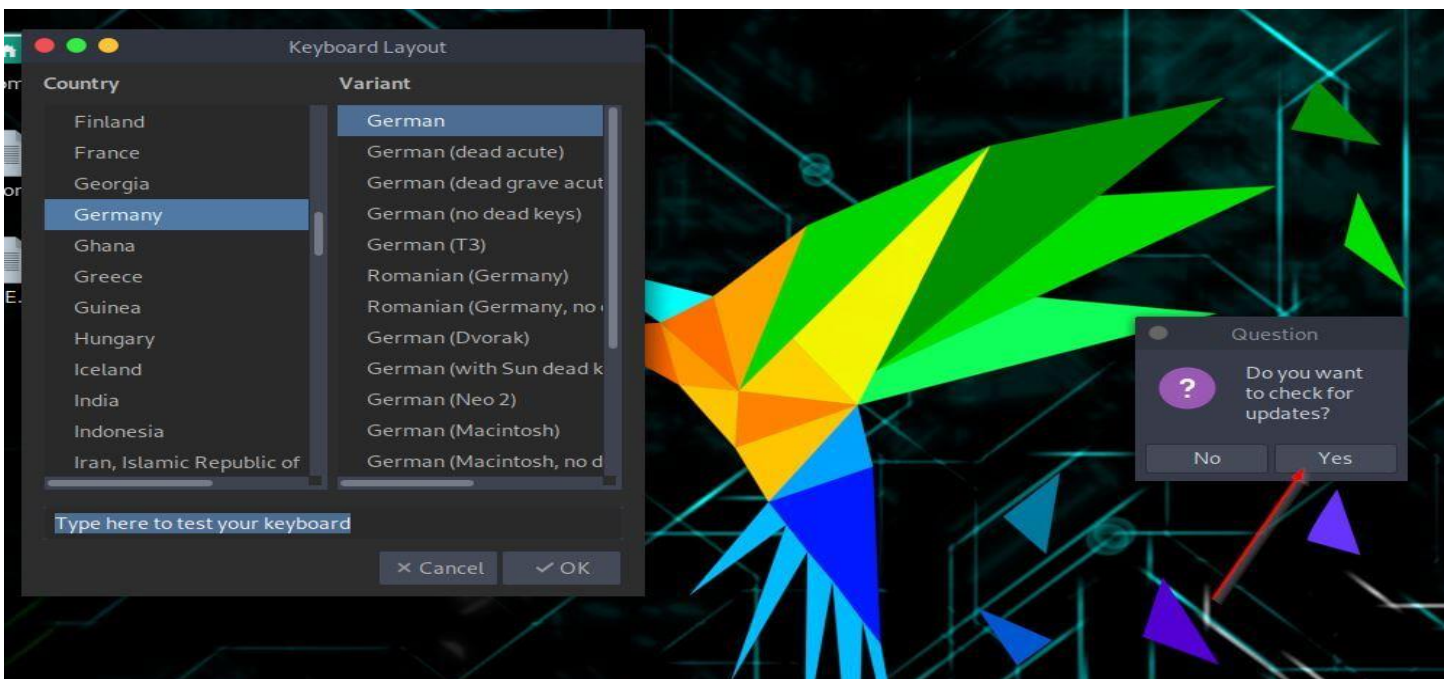
WEEK 1

INSTALLATION





UPDATING THE APPLIANCE



Week 1 — Installation & Updating the Appliance The first week focused on setting up the working environment by installing the necessary tools and platform. Keeping the appliance updated ensures the latest security patches and tool versions are in place. A properly configured environment is the foundation for all subsequent ethical hacking activities.

WEEK 2

WEB VULNERABILITY SCANNING

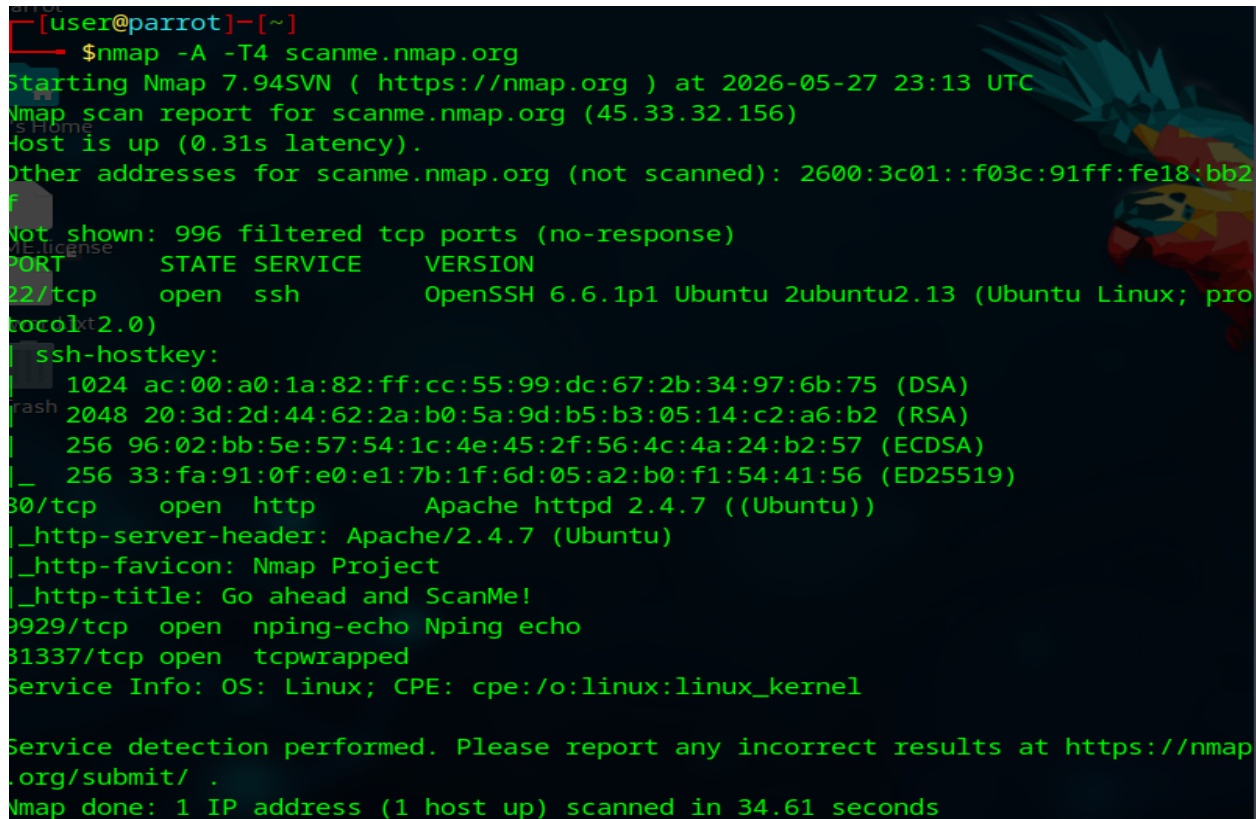
```
the http server's resources causing Denial Of Service.
Disclosure date: 2009-09-17
References:
  https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
  http://ha.ckers.org/slowloris/
http-iis-webdav-vuln: WebDAV is DISABLED. Server is not currently vulnerable.
http-stored-xss: Couldn't find any stored XSS vulnerabilities.
http-csrf: Couldn't find any CSRF vulnerabilities.
http-fileupload-exploiter:
  Couldn't find a file-type field.
  Couldn't find a file-type field.
  Couldn't find a file-type field.
http-enum:
  /login.html: Possible admin folder
3517/tcp open  802-11-iapp

NSE: Script Post-scanning.
Initiating NSE at 00:05
Completed NSE at 00:05, 0.00s elapsed
Initiating NSE at 00:05
Completed NSE at 00:05, 0.00s elapsed
Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 187.78 seconds
[user@parrot]-[~]
$
```


Week 2 — Web Vulnerability Scanning The second week involved actively scanning web targets to identify potential security weaknesses and misconfigurations. This phase highlights how automated tools can quickly surface exploitable entry points in a web application. Understanding scan results is critical for distinguishing real vulnerabilities from false positives.

WEEK 3

DNS FOOTPRINTING

A terminal window showing an Nmap scan of scanme.nmap.org. The output includes the Nmap version (7.94SVN), the target IP (45.33.32.156), and a list of open ports and services. The scan was performed on 2026-05-27 at 23:13 UTC. The terminal background features a parrot illustration.

```
[user@parrot]~$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-27 23:13 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.31s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256  96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
9929/tcp  open  nping-echo   Nping echo
81337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.61 seconds
```

PORT SCANNING

```
--[x]--[user@parrot]--[~]
$ nmap -sVC -p1-100 192.168.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-27 23:46 UTC
WARNING: Service 192.168.1.1:80 had already soft-matched rtsp, but now soft-matched sip; ignoring second value
Nmap scan report for 192.168.1.1
Host is up (0.0042s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  domain Cloudflare public DNS
8080/tcp   open  rtsp
  _rtsp-methods: ERROR: Script execution failed (use -d to debug)
  _http-title: Router
  fingerprint-strings:
    GetRequest:
      HTTP/1.0 200 Ok
      Date: Wed, 27 May 2026 23:46:09 GMT
      Cache-Control: no-cache,no-store
      X-Frame-Options: SAMEORIGIN
      X-XSS-Protection: 1; mode=block
      X-Content-Type-Options: nosniff
      Content-Type: text/html; charset=UTF-8
      Content-Length: 5858
      Last-Modified: Thu, 17 Apr 2025 07:53:02 GMT
      Connection: close
      <!--
      @,@Author: ,: your name
      @,@Date: ,: 2020-12-15 13:56:49
      @,@LastEditTime: ,: 2021-08-12 18:32:42
      @,@LastEditors: ,: Please set LastEditors
```

Week 3 — DNS Footprinting, Port Scanning & Motivation Week three expanded reconnaissance to include DNS enumeration and port scanning, mapping out the attack surface of a target. The motivation section reinforced that every exposed record, port, or email pattern represents a real security risk. Key lessons drawn include using CDNs to hide origin IPs, restricting zone transfers, and closing unnecessary ports.

MOTIVATION

Every open port, DNS record, email pattern, and exposed technology reveals a potential attack surface, proving that obscurity is not security. Key lessons include hiding origin IPs with CDNs, restricting DNS zone transfers, using unpredictable email formats, closing unnecessary ports, and removing version-disclosing headers. However, major challenges were avoiding false positives in automated vulnerability scans, distinguishing active subdomains from stale DNS entries, and consistently securing all assets without breaking legitimate services. Additionally, ethical testing requires careful scoping to prevent unintended disruption, while defenders must balance usability with strict error-message standardization to block user enumeration.

WEEK 4

COMPANY & DOMAIN INFORMATION

I. IP ADDRESS & HOSTING

```
[user@parrot]~$ ping clarity-ventures.com
PING clarity-ventures.com (104.20.9.104) 56(84) bytes of data:
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=1 ttl=255 time=50.4 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=2 ttl=255 time=44.8 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=3 ttl=255 time=51.0 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=4 ttl=255 time=50.1 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=5 ttl=255 time=48.3 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=6 ttl=255 time=46.0 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=7 ttl=255 time=54.0 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=8 ttl=255 time=41.6 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=9 ttl=255 time=50.9 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=10 ttl=255 time=48.3 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=11 ttl=255 time=36.6 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=12 ttl=255 time=45.2 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=13 ttl=255 time=52.0 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=14 ttl=255 time=50.1 ms
64 bytes from 104.20.9.104 (104.20.9.104): icmp_seq=15 ttl=255 time=49.3 ms
^C
--- clarity-ventures.com ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14042ms
rtt min/avg/max/mdev = 36.632/47.911/54.036/4.298 ms
```



```

[user@parrot]-[~]
$nslookup clarity-ventures.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   clarity-ventures.com
Address: 104.20.9.104
Name:   clarity-ventures.com
Address: 104.20.10.104
Name:   clarity-ventures.com
Address: 2606:4700:10::6814:968
Name:   clarity-ventures.com
Address: 2606:4700:10::6814:a68

```

EMAIL PATTERNS

```

[user@parrot]-[~]
$host -t mx clarity-ventures.com
clarity-ventures.com mail is handled by 10 mail.clarity-ventures.com.

[user@parrot]-[~]
$nmmap -p 25 --script smtp-enum-users mail.clarity-ventures.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-03 22:22 UTC
Nmap scan report for mail.clarity-ventures.com (206.127.30.140)
Host is up (0.42s latency).
rDNS record for 206.127.30.140: 206-127-30-140.fwd.datafoundry.com

PORT      STATE SERVICE
25/tcp    open  smtp
| smtp-enum-users:
|_ SMTP EHLO mail.clarity-ventures.com: failed to receive data: failed to receive data

Nmap done: 1 IP address (1 host up) scanned in 2.99 seconds

```

PORT SCANNING

```
(user@parrot)-[~]  
$nmap -Pn clarity-ventures.com  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-03 22:30 UTC  
Nmap scan report for clarity-ventures.com (104.20.10.104)  
Host is up (0.074s latency).  
Other addresses for clarity-ventures.com (not scanned): 2606:4700:10::6814:968  
Not shown: 996 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp    open  https  
8080/tcp   open  http-proxy  
8443/tcp   open  https-alt  
  
Nmap done: 1 IP address (1 host up) scanned in 10.43 seconds
```

DISCOVERING SUBDOMAINS

```
[x]-[user@parrot]-[~]
$ nmap --script dns-brute clarity-ventures.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-03 23:13 UTC
Nmap scan report for clarity-ventures.com (104.20.10.104)
Host is up (0.059s latency).
Other addresses for clarity-ventures.com (not scanned): 2606:4700:10::6814:968
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp  open  http-proxy
8443/tcp  open  https-alt

Host script results:
| dns-brute:
|   DNS Brute-force hostnames:
|     admin.clarity-ventures.com - 104.20.10.104
|     admin.clarity-ventures.com - 104.20.9.104
|     admin.clarity-ventures.com - 2606:4700:10::6814:968
|     admin.clarity-ventures.com - 2606:4700:10::6814:a68
|     images.clarity-ventures.com - 104.20.10.104
|     images.clarity-ventures.com - 104.20.9.104
|     images.clarity-ventures.com - 2606:4700:10::6814:968
|     images.clarity-ventures.com - 2606:4700:10::6814:a68
|     direct.clarity-ventures.com - 207.207.33.175
|     info.clarity-ventures.com - 199.60.103.225
|     info.clarity-ventures.com - 199.60.103.31
|     info.clarity-ventures.com - 2606:2c40::c73c:671f
|     info.clarity-ventures.com - 2606:2c40::c73c:67e1
|     web.clarity-ventures.com - 52.159.97.111
|     cdn.clarity-ventures.com - 104.20.10.104
|     cdn.clarity-ventures.com - 104.20.9.104
|     mail.clarity-ventures.com - 206.127.30.140
|     cdn.clarity-ventures.com - 2606:4700:10::6814:968
|     cdn.clarity-ventures.com - 2606:4700:10::6814:a68
|     ftp.clarity-ventures.com - 207.207.33.175
|     www.clarity-ventures.com - 104.20.10.104
|     www.clarity-ventures.com - 104.20.9.104
|     www.clarity-ventures.com - 2606:4700:10::6814:968
|     www.clarity-ventures.com - 2606:4700:10::6814:a68
|     smtp.clarity-ventures.com - 92.204.80.20
|     help.clarity-ventures.com - 199.60.103.225
|     help.clarity-ventures.com - 199.60.103.31
|     help.clarity-ventures.com - 2606:2c40::c73c:671f
|     help.clarity-ventures.com - 2606:2c40::c73c:67e1
|     dev.clarity-ventures.com - 71.41.147.175
|     stage.clarity-ventures.com - 50.28.62.94
Nmap done: 1 IP address (1 host up) scanned in 12.07 seconds
```

DETECTING TECHNOLOGIES & SERVICES

```
[user@parrot]-[~]
$ nmap -sV clarity-ventures.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-04 00:00 UTC
Nmap scan report for clarity-ventures.com (104.20.9.104)
Host is up (0.098s latency).
Other addresses for clarity-ventures.com (not scanned): 2606:4700:10::6814:968
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Cloudflare http proxy
443/tcp   open  ssl/http Cloudflare http proxy
8080/tcp  open  http    Cloudflare http proxy
8443/tcp  open  ssl/http Cloudflare http proxy

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.38 seconds
```

ENUMERATE USERS

```
[user@parrot]-[~]
$ nmap -p 25 --script smtp-enum-users --script-args smtp-enum-users.methods={VRFY} 104.20.9.104
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-04 01:33 UTC
Nmap scan report for 104.20.9.104
Host is up (0.080s latency).

PORT      STATE SERVICE
25/tcp    filtered smtp

Nmap done: 1 IP address (1 host up) scanned in 1.04 seconds
```

Week 4 — Company & Domain Analysis, Security Risks The final week brought all recon techniques together — IP lookup, email patterns, subdomain discovery, technology fingerprinting, and user enumeration — applied to a real company domain. Each finding was tied to a concrete security risk, from phishing enablement to DDoS targeting and brute-force attacks. The week concluded with a sobering reminder that the recon document itself, if leaked, becomes a ready-made attack blueprint.

SECURITY RISKS

Exposed IP & Hosting Information

Risk: Attackers can pinpoint the hosting provider and IP ranges, making DDoS attacks, infrastructure mapping, or exploiting known vulnerabilities in that host easier.

Email Pattern Discovery

Risk: Revealing email formats (e.g., first.last@company.com) enables targeted phishing, social engineering, or credential stuffing attacks.

Open Ports & Services

Risk: Unnecessary or insecure open ports (e.g., SSH, RDP, outdated database ports) increase the attack surface, potentially allowing remote access, lateral movement, or exploitation of unpatched services.

Subdomain Enumeration

Risk: Forgotten, unmaintained, or poorly secured subdomains (like dev, test, or admin panels) may expose internal apps, staging data, or APIs to external attackers.

Technology & Service Fingerprinting

Risk: Knowing specific software stacks (web servers, CMS, frameworks, libraries) lets attackers search for known CVEs or default configurations to exploit.

User Enumeration via Nmap Scripts

Risk: Attackers can validate valid usernames (e.g., from SMTP, RDP, or SMB services), which dramatically increases the success rate of brute-force or password spraying attacks.

Internal Reconnaissance Data Leakage

Risk: The very existence of this document (combining multiple recon outputs) suggests sensitive network mapping data has been collected and stored without adequate protection — if leaked, it becomes a ready-made attack blueprint.